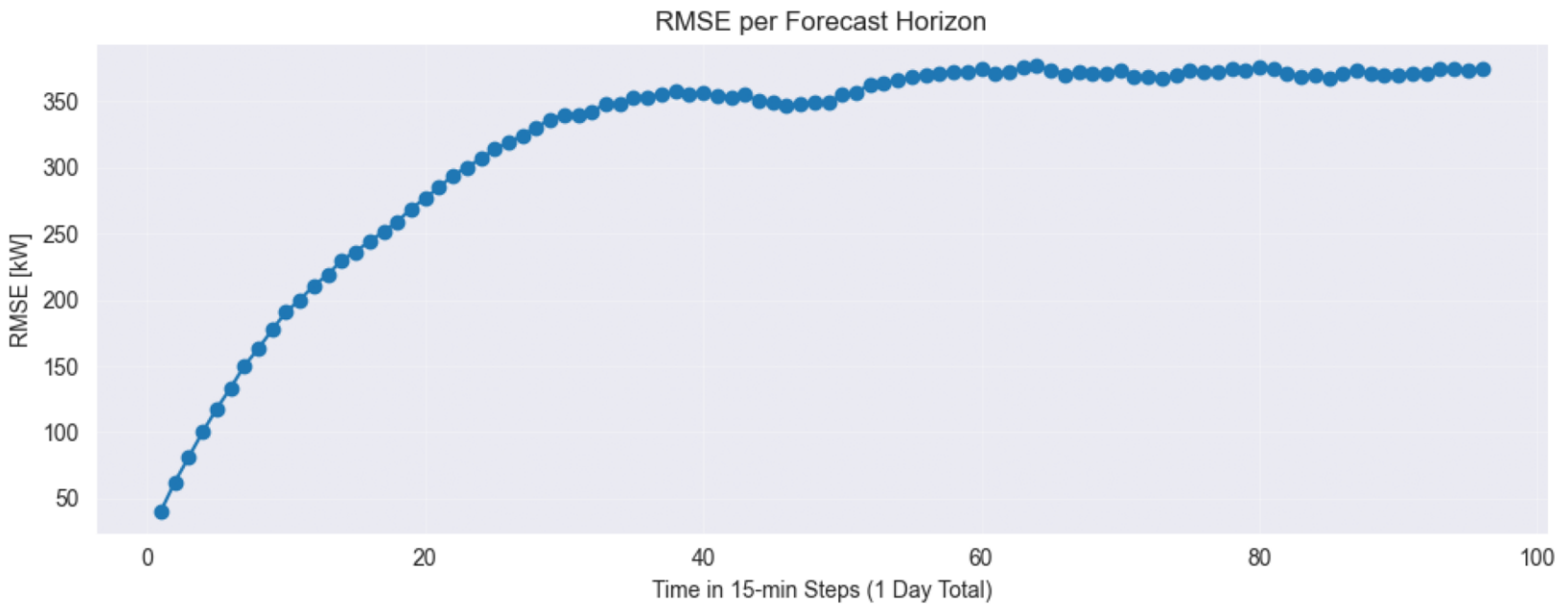
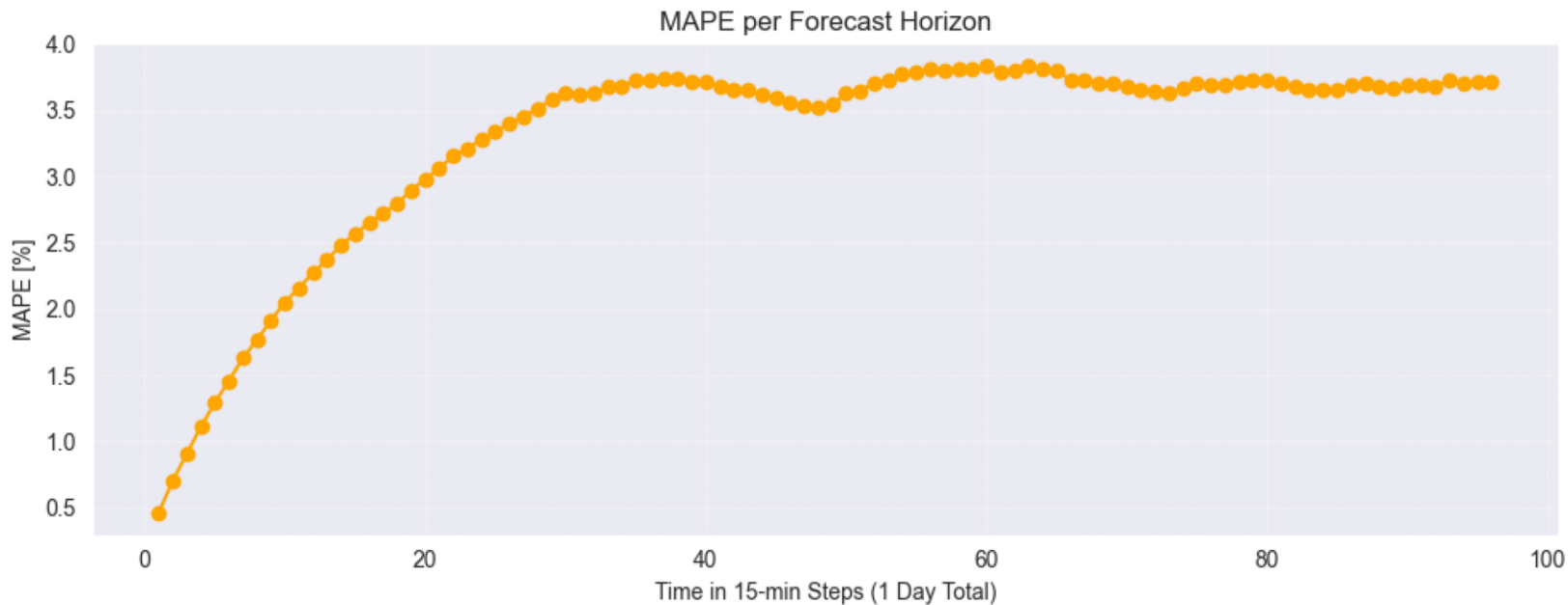
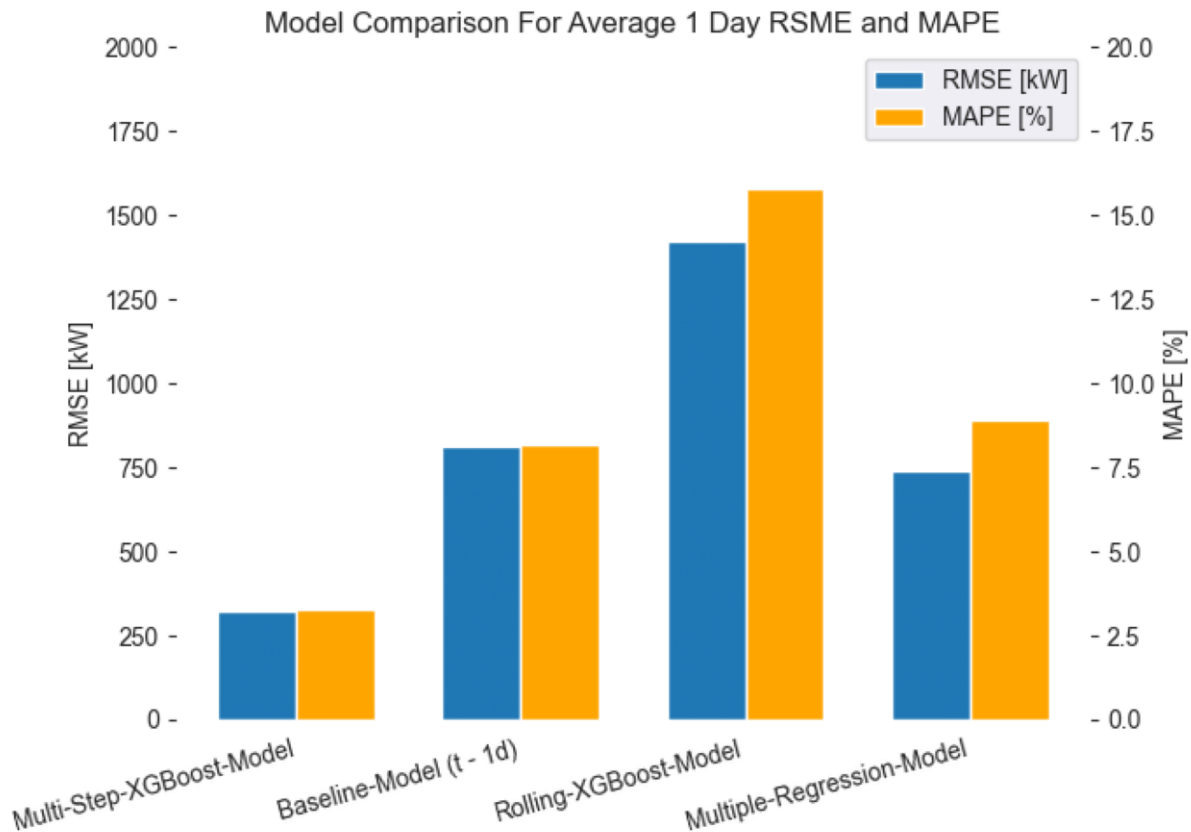




- Error rises steeply for early horizons, then plateaus after ~15 hours or 60 steps
 - Daily RMSE (Root Mean Square Error) stabilises ≈ 360 kW
 - Indicates diminishing—but stable—accuracy with longer horizons

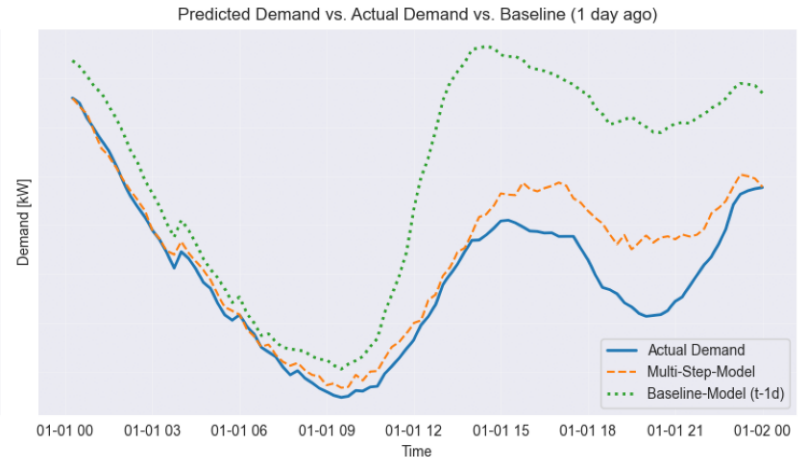
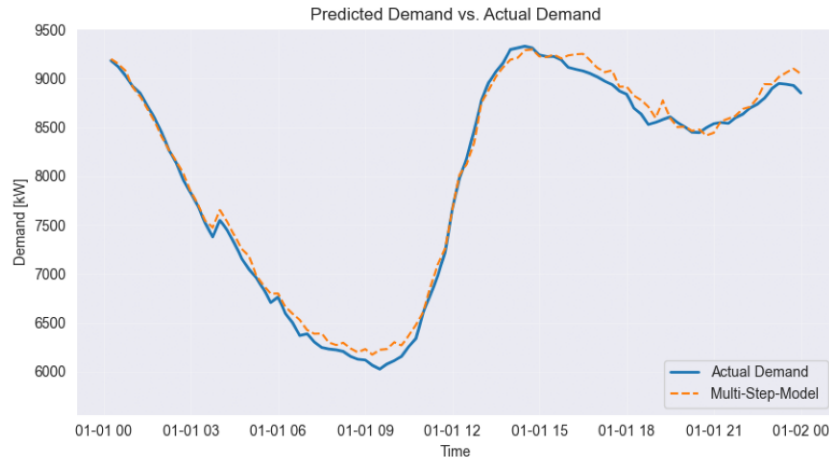


- Relative error climbs to $\sim 3.8\%$ and levels off
- MAPE (Mean Absolute Percentage Error) stays $< 4\%$ across the full 24-h horizon, confirming robustness
 - Trend mirrors RMSE curve



- Multi-Step XGBoost achieves the lowest RMSE & MAPE of all models
 - Rolling & Multiple-Regression models show 2-4× higher errors
 - Baseline (t-1 d) lags far behind the Multi-Step model
 - Rolling performs worst, likely due to error compounding across steps
- Model forecasts future steps based on earlier (possibly wrong) predictions

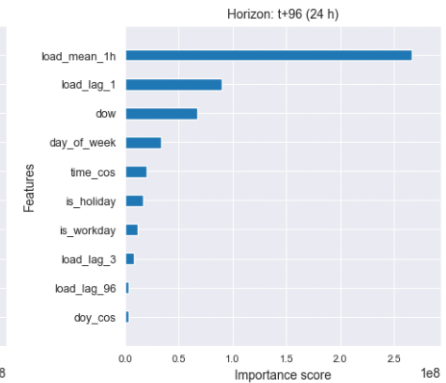
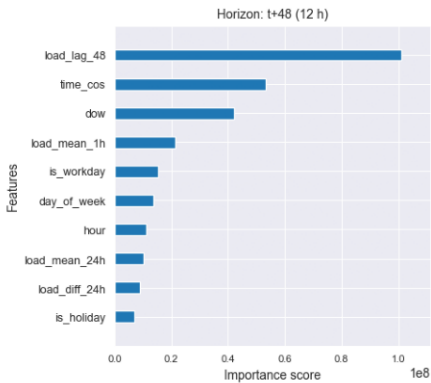
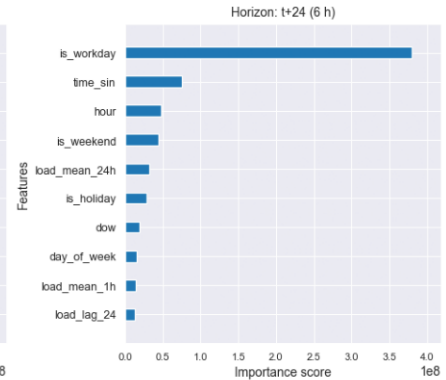
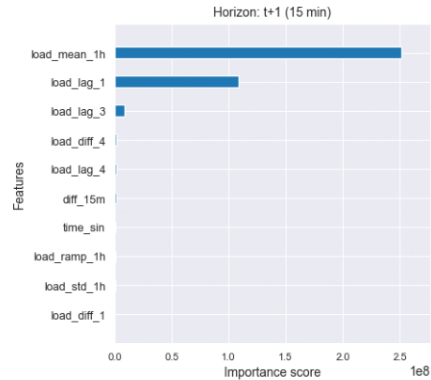
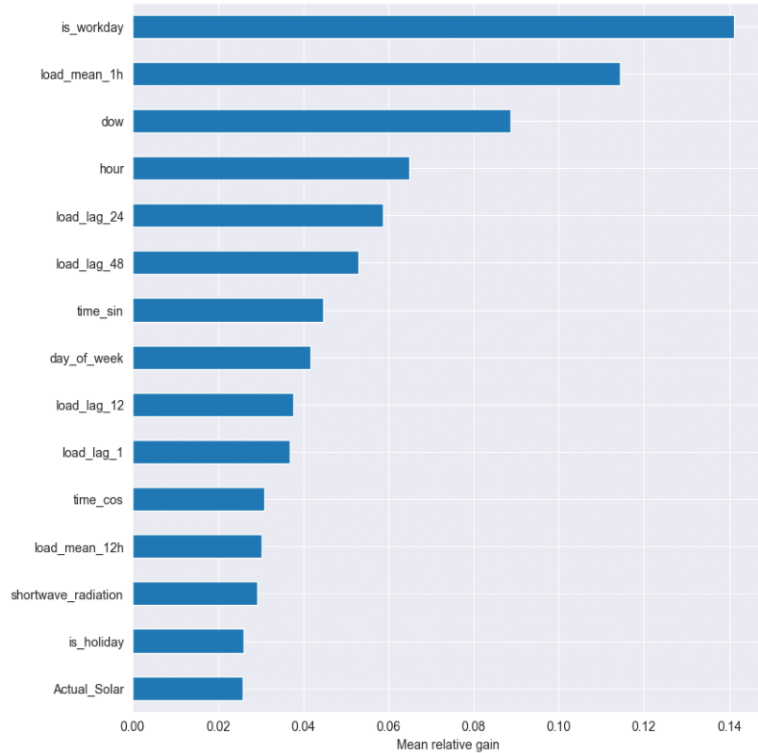
24-h Forecast



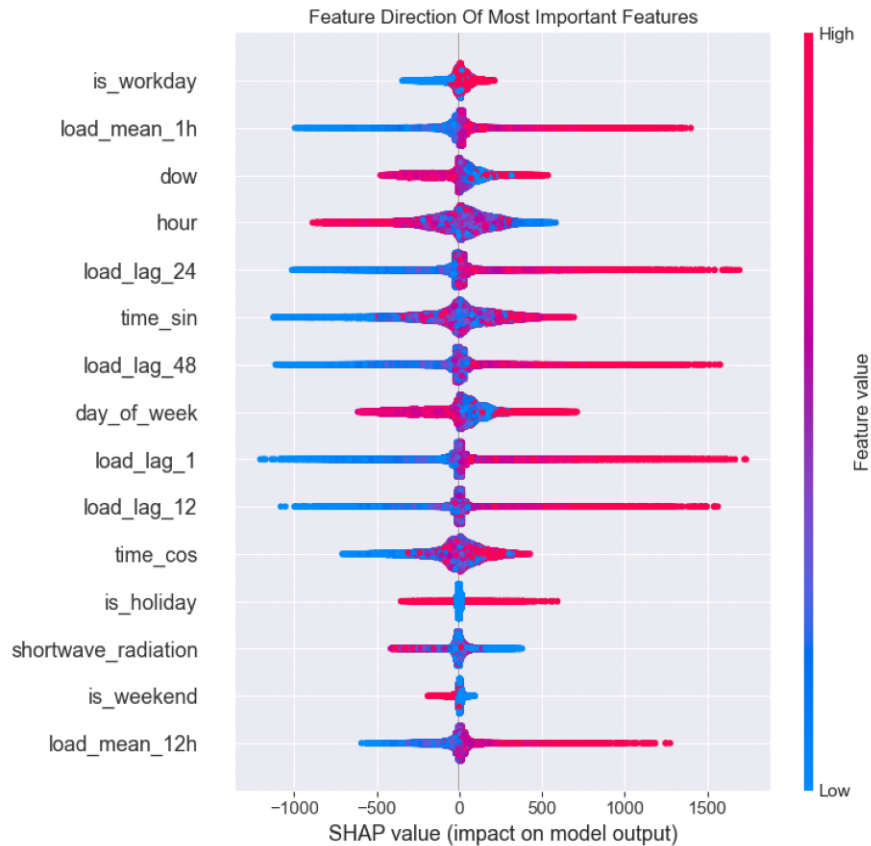
- Multi-Step forecast tracks actual demand closely across the day
- Baseline systematically over/under-estimates peaks & troughs
- Visual evidence of superior temporal alignment for Multi-Step

XGBoost Feature Importance Across Forecast Horizons

Overall XGBoost Feature Importance
(Average over 96 horizons)

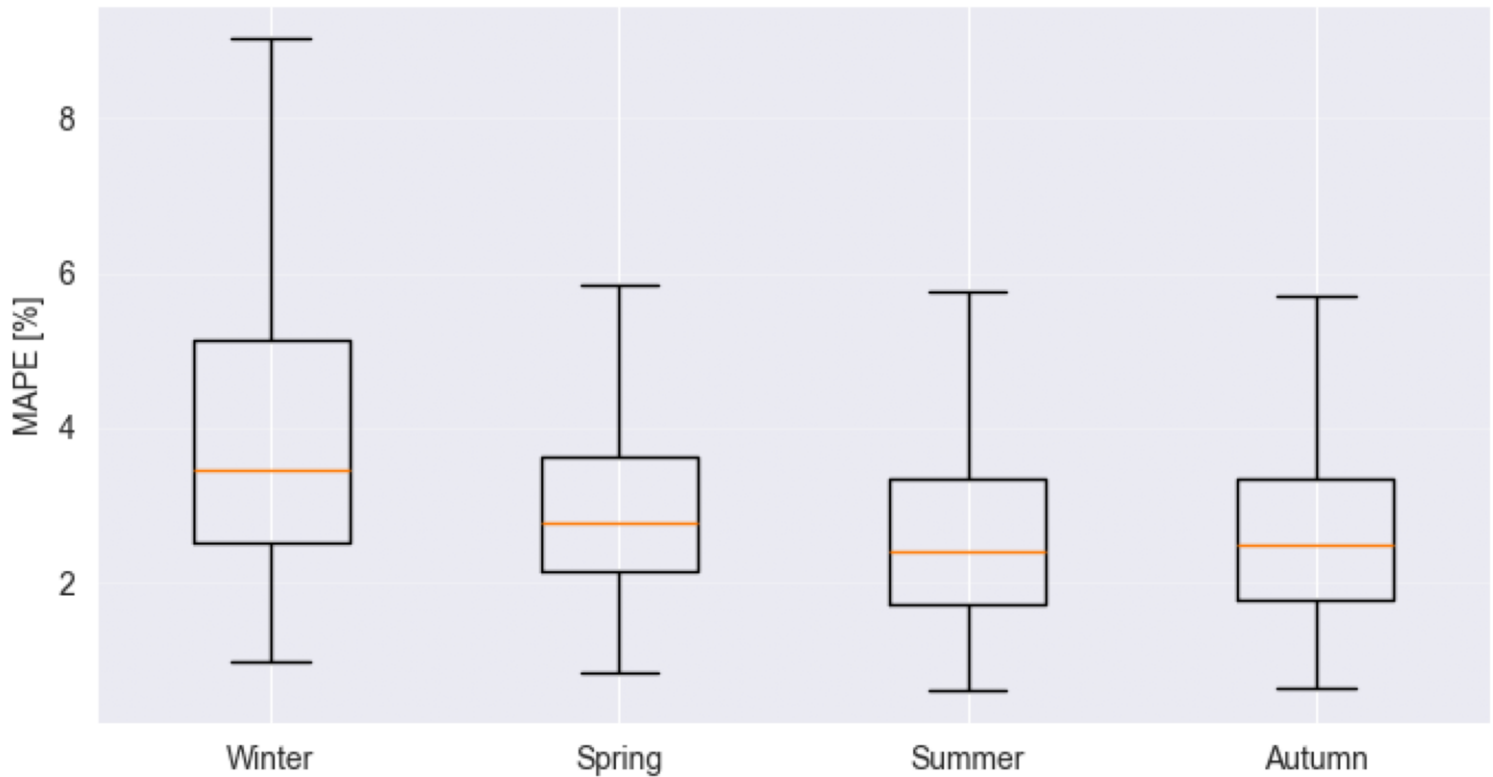


- Workday flag & 1-h mean load are most informative overall
- Lagged load dominates short horizons; calendar features dominate mid-range
 - Feature importance profile shifts with forecast horizon



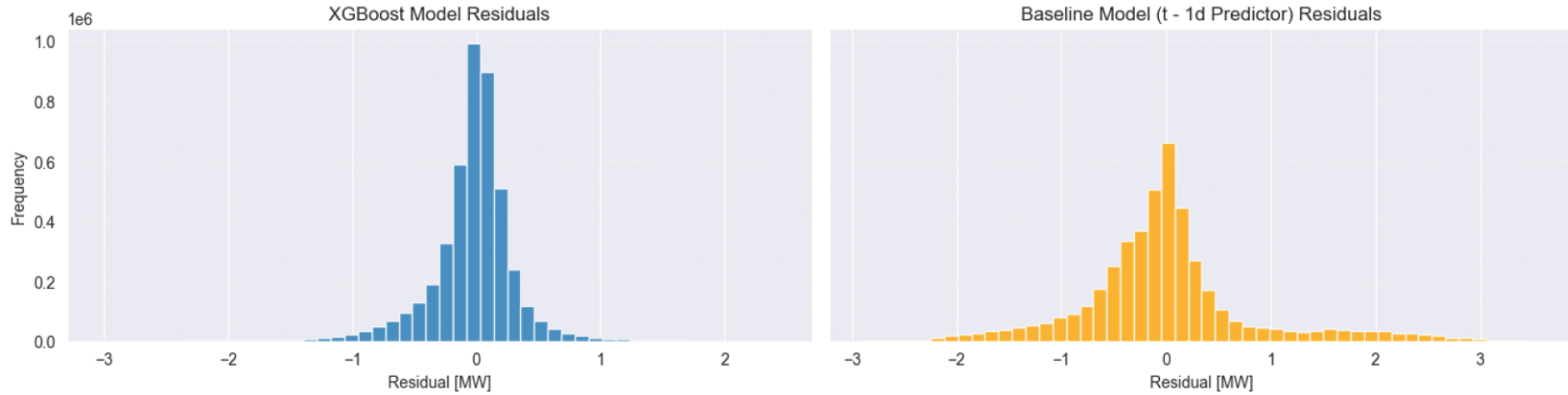
- Is_workday indicates higher demand on weekdays compared to weekends
- Shortwave Radiation contributes negatively, likely due to reduced heating and self-generation
 - Recent load and time-based features show clear directional influence on predictions

Seasonal Error Analysis For 1 Day-Ahead MAPE



- Winter exhibits highest median error & widest spread (~3-4 %)
 - Spring-Autumn medians ~2-2.5 % with tighter IQR
- Slightly lower accuracy in winter—likely heating-driven variability

Residual Distribution - Test Year 2024



- Multi-Step residuals tightly centred near 0 MW (narrow spread)
 - Baseline residuals broader, with more extreme deviations
 - Confirms reduced bias & variance of the Multi-Step model